Douglas Gordon Daly

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SUMMARY

Principal AI Engineer and Technical Leader with 20+ years of experience building systems that help organizations understand what is happening, why it is happening, and what to do next. Expertise spans operational intelligence, decision-support systems, attribution analytics, observability, graph analytics, and AI-enabled business systems.

Designed and delivered solutions ranging from a patented radar resource-management system and enterprise-scale operational monitoring platforms to merchant intelligence systems, reliability analytics, graph-based dependency modeling, GenAI assistants, and AI-enabled decision support solutions. Combines deep expertise in machine learning, cloud platforms, data engineering, and modern LLM architectures with a proven ability to translate complex business problems into practical, measurable outcomes.

CORE TECHNOLOGIES

AI & Machine Learning: LLMs | RAG | Agentic AI | LangGraph | CrewAI | AutoGen | MCP | Prompt Engineering | PEFT/LoRA | Fine-Tuning | Diffusion Models | CLIP | Scikit-Learn | XGBoost
Cloud & Data Platforms: AWS | GCP | Vertex AI | BigQuery | SageMaker | Bedrock | S3 | Neptune
Programming & Analytics: Python | SQL | Tableau | Pandas | NumPy | Jupyter | Elastic Stack | Splunk | Kafka
Data & Systems Engineering: Graph Analytics | Observability | Operational Intelligence | Entity Resolution | Data Modeling | Forecasting | Attribution Analytics

PROFESSIONAL EXPERIENCE

Principal Consultant | Daly Engineers LLC | Jan 2026 - Present

Consolidated Edison (via ABC Consulting)

- Led analytical readiness assessment for Oracle/EBS and Maximo data migrated to BigQuery, establishing the foundation for cross-domain analytics across construction operations and financial systems.
- Developed automated profiling, type inference, duplicate detection, candidate key analysis, and cross-system join validation frameworks across 100+ source tables and millions of operational and financial records.
- Developed a phased analytical maturity roadmap adopted as the foundation for anomaly detection, construction analytics, and AI-enabled decision support initiatives.
- Built a governed analytics assistant using Gemini, BigQuery, and tool-calling architectures that applied analyst rules and operational controls while maintaining predictable and auditable execution.

NBCUniversal (via Apex Systems)

- Built AWS-based AI and graph analytics pipelines using SageMaker, Terraform, S3, and containerized workloads.
- Developed graph-ready service dependency datasets from API telemetry and operational metrics to support digital twin initiatives.
- Evaluated graph analytics and graph machine learning approaches using Neptune, GraphStorm, Bedrock, and LangGraph for dependency analysis and operational intelligence.
- Authored architecture and analytical readiness assessments identifying data and operational gaps affecting long-term digital twin objectives.

AI Instructor | SimpliLearn | October 2025 - Present

- Delivered professional AI training across multi-month cohorts for software engineers, data scientists, and technical professionals through university- and enterprise-sponsored continuing education programs.
- Taught transformer architectures, LLMs, RAG, agentic AI, multimodal AI, MCP, diffusion models, and enterprise AI governance.
- Developed hands-on demonstrations, Jupyter notebooks, and end-to-end applications covering modern AI workflows, evaluation techniques, and deployment considerations.
- Mentored students on AI architecture, governance, implementation tradeoffs, and enterprise adoption strategies.

AI Instructor | Interview Kickstart | March 2025 - Present

- Delivered AI and machine learning instruction for software engineering professionals pursuing advanced technical roles.
- Taught practical applications of LLMs, RAG, agentic AI, and modern AI development workflows.
- Mentored students on AI architecture, implementation patterns, and technology selection tradeoffs.

Data Scientist | Intuit (via The Cydio Group) | Apr 2025 - Jul 2025

- Delivered analytics supporting business and operational decision-making for QuickBooks Capital.
- Investigated attribution anomalies across customer acquisition funnels, marketing campaigns, and conversion pathways to improve measurement accuracy and identify emerging business trends.
- Consolidated fragmented reporting into unified Tableau dashboards and automated executive reporting workflows using Python and SQL, reducing report preparation from days to hours.
- Partnered with finance, marketing, and product stakeholders to translate attribution and performance data into actionable business recommendations.

Data Scientist | Meta | October 2021 - January 2025

- Defined reliability KPIs and measurement frameworks for Meta FinTech, establishing leading indicators used to guide reliability planning, prioritization, and performance management.
- Developed forecasting and performance-management frameworks across business pillars and sub-pillars, translating engineering initiatives into projected reliability outcomes used for planning, goal setting, and leadership decision-making.

- Built attribution frameworks linking incidents, alerts, defects, and operational metrics to engineering organizations and service ownership structures, enabling more accurate accountability and performance measurement.
- Designed and developed the ReMon framework, enabling engineering teams to build comprehensive monitoring and alerting systems without requiring expertise in complex internal tooling; adopted across Meta FinTech, contributing to the automatic detection of more than 15 SEVs in 2024.
- Delivered impact sizing, ROI, and risk exposure analyses supporting investment and prioritization decisions across platform reliability, privacy compliance, authentication, testing infrastructure, and data platform initiatives.

Data Scientist | Samsung Electronics (via Harvey Nash) | Dec 2020 - Oct 2021

- Developed and deployed gradient-boosting propensity models used to target digital marketing campaigns, increasing campaign conversions by 146% and campaign-attributed revenue by 229%.
- Built customer sentiment analysis pipelines using NLTK and survey data to identify product and customer experience insights.
- Automated audience segmentation and campaign targeting workflows, improving efficiency and scalability across digital marketing initiatives.
- Improved marketing attribution and performance measurement capabilities through predictive modeling and analytics-driven decision support.

Head of Data Science | Facteus | Oct 2017 - Nov 2020

- Led data science initiatives on a 30B+ transaction dataset, creating analytical products and market-intelligence capabilities through large-scale feature engineering, merchant intelligence, and consumer-spending analytics.
- Developed merchant, brand, industry, and storefront resolution methodologies that transformed cryptic payment-card transactions into business-identifiable entities, enabling accurate market intelligence across more than 1,700 businesses with greater than 99% transaction coverage.
- Evaluated prospective data partnerships by assessing signal quality, coverage, and business value, supporting strategic investment and acquisition decisions.
- Produced transaction-based industry analyses during the COVID-19 pandemic, identifying shifts in consumer behavior, regional economic impacts, and e-commerce adoption trends before traditional reporting sources became available.
- Created data-driven reports and visualizations used by hedge funds, partners, and internal stakeholders to understand emerging market trends and business performance.

Expert SEO Solutions Architect | Nike (via Intersoft Inc.) | Nov 2016 - Sep 2017

- Designed a scalable SEO and information-architecture framework for Nike's product discovery experience, balancing customer navigation paths with search-engine indexation requirements across a rapidly expanding product catalog.
- Developed canonical URL, redirect, and XML sitemap strategies that improved search visibility while preventing duplicate content, redirect conflicts, and circular routing issues.
- Built Node.js proof-of-concept solutions demonstrating the feasibility and effectiveness of proposed SEO approaches, supporting adoption by business and engineering teams.

Senior Manager, Operations Analysis | Capital One | Jan 2012 - Dec 2016

- Developed predictive capacity-management models for customer-facing banking platforms, forecasting peak demand scenarios using customer-demand, temporal, and business-process drivers to support infrastructure planning and risk management.
- Identified critical gaps in enterprise monitoring operations after discovering that major incidents were frequently detected by customer call channels rather than operational monitoring systems.
- Conceived, designed, and led adoption of RITHM (Real-Time IT Health Monitoring), an operational intelligence platform that aggregated monitoring signals by business capability and system ownership, reducing major incident detection times from hours to minutes.
- Secured executive sponsorship for enterprise monitoring modernization, expanding initial Splunk-based prototypes into a scalable Kafka and Elastic Stack solution aligned with Capital One's open-source strategy.
- Partnered with CTO-level leadership to define enterprise monitoring strategy, operational health metrics, and reliability-improvement initiatives.
- Awarded the Capital One Business Innovation Award for contributions to operational intelligence and enterprise monitoring transformation; presented work at internal and external industry conferences.

Senior Principal Systems Engineer / Program Manager | Raytheon | Mar 2005 - Dec 2011

- Led systems engineering organizations of 40-50 engineers specializing in modeling, simulation, analysis, and advanced aerospace systems, providing technical leadership, career development, and proposal support for programs valued up to $50M.
- Turned around an over-budget classified fixed-price technology program at approximately one-third completion by establishing execution plans, aligning engineering teams, and proactively managing technical and program risks; delivered the program under budget and nearly doubled the original profit target on a historically challenging delivery effort.
- Served as Principal Investigator for advanced signal-generation and radar resource-management algorithms, leading research, simulation, flight testing, and operational deployment across multiple tactical and surveillance aircraft programs.
- Developed a patented fuzzy-logic resource-management system (U.S. Patent, 2014) resulting from radar scheduling innovations deployed on major Department of Defense aircraft platforms.

SELECTED PROJECTS

GraphRAG Engine & LoRA Coverage Assistant | GitHub | 2025

- Built a GraphRAG system for multi-hop benefits reasoning with fully traceable retrieval paths across entities, clauses, and supporting evidence.
- Fine-tuned a LoRA adapter on a 7B open-source LLM to enforce a strict JSON schema (decision/reason/disclaimer) for reliable downstream automation.

Agent-to-Agent Protocol MVPs | GitHub | 2026

- Agent Olympics: built a small evaluation runner that benchmarks a generalist agent vs discovered specialists across three task categories with explicit scoring criteria and summary reporting.

- AI PEMDAS: built an instrumented coordination workflow where an evaluator routes sub-steps to ADD vs MULT agents and logs calls for end-to-end traceability (FastAPI JSON-RPC).

Agentic Multimodal Orchestrator | GitHub | 2025

- Orchestrated structured retrieval, layout planning, and diffusion prompting to generate data-driven posters/maps with repeatable, template-based workflows.
- Implemented template constraints and parameterized prompts to ensure consistency across runs.

EDUCATION

MBA | (UCLA Anderson School of Management): Top Honors | GPA: 3.9
M.S., Applied Statistics | (Rochester Institute of Technology): GPA: 3.9
M.S., Electrical Engineering | (University of Washington): GPA: 3.7
B.S., Physics (Minor: Mathematics) | (Humboldt State University): GPA: 3.4

PATENTS & RECOGNITION

- Lead Inventor, U.S. Patent #8635622: Method and System for Resource Management Using Fuzzy Logic Timeline Filling (2014)
- Capital One Business Technology Innovation Award (2014)
- Conference Chair & Speaker - Customer Analytics Innovation Summit (2016)